W-S-564C
April 10, 1991
SUPERSEDING
W-S-564B
October 15, 1985

FEDERAL SPECIFICATION

SOLDERING GUNS, ELECTRIC

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric soldering guns and replacement tips and heating elements. Soldering guns are generally used for intermittent soldering tasks where rapid availability for use is required without the continuous consumption of electric current.

1.2 Classification. This specification covers soldering guns of the following types, classes, sizes, and temperatures as specified. The type, class, size, temperature, and quantity to be furnished shall be as specified (see 6.2.1).

Type I - Wire tip, with transformer.

Size 1 - Heat volume less than 85 seconds.
Size 2 - Heat volume less than 65 seconds.
Size 3 - Heat volume less than 45 seconds.

Type II - Solid tip.

Type IV - Solid tip, temperature controlled.

Class 1 - Fixed temperature.

Temperature 600 - 600 degrees Fahrenheit (tip temperature).
Temperature 700 - 700 degrees Fahrenheit (tip temperature).

Class 2 - Variable temperature.

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL STANDARDS

FED-STD-H28 - Screw Thread Standards for Federal Services.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

MILITARY SPECIFICATION

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

MILITARY STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packaging, Standard Practice For.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

UNDERWRITERS LABORATORY (UL) STANDARDS FOR SAFETY

UL 499 - Electric Heating Appliances.

(Application for copies should be addressed to Testing and Materials, 1916 Race St., Underwriters Laboratories, Inc., 333 Pfingsten Rd. Northbrook, IL 60062-2096.)

3. REQUIREMENTS

3.1 Material. The material used in the construction of the soldering guns shall be as specified herein. When material is not specified it shall conform to a high quality commercial grade.

3.1.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent possible.

3.1.2 Fire and casualty hazards. Each bidder shall submit to the contracting agency proof that the soldering guns they propose to supply under this specification conform to the applicable requirements of UL-499. The label or listing of the Underwriters Laboratories, Inc. may be accepted as evidence that the soldering guns conform to this requirement. Compliance with UL-499 does not absolve the supplier from complete compliance with the requirements of this specification in order to secure acceptance of their product.

3.2 First article. When specified (see 6.2.1), one sample of each item shall be submitted for first article inspection (see 4.3 and 6.3). The first article may be a preproduction model or an initial production model.

3.3 General Requirements.

3.3.1 Design. The soldering guns shall be new and one of the manufacturers current models that will meet the requirements of this specification. The soldering guns shall be furnished complete and ready for use.

3.3.2 Workmanship. Workmanship of the soldering guns shall be of a quality equal to that of the comparable commercial products furnished to the commercial market.

3.3.3 Construction. The soldering guns shall be constructed of parts which are new, and free of defects. The soldering guns shall be so constructed that all parts, which are subject to replacement during the normal life of the guns, are easily removable (for replacement) without the necessity of special tools.

3.3.4 Threads. All fasteners used on the soldering gun shall conform to FED-STD-H28.

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3.3.5 Finish. Unless constructed of corrosion resistant material, all metal parts of the soldering guns, other than soldering tips, shall be suitably protected to prevent corrosion from soldering flux and fumes.

3.3.6 Identification marking. Each soldering gun shall be marked in a permanent manner with the following:

a. The manufacturer's name or with a trademark of such known character that source of manufacture may be readily determined.

b. Manufacturer's model or part number.

3.3.7 Impact resistance. The soldering guns shall withstand a 3 foot drop on a hard surface without damage that would cause a safety hazard to the user.

3.3.8 Heat Volume. The heat volume of the type I and type II soldering guns shall be defined as the time required for the soldering gun, when energized and with the tip placed at one end of a 0.125 by 0.5 by 1.5 inch copper bar, to melt 0.0625 inch diameter 60/40 tin-lead solder placed on the opposite end of the bar. The heat volume shall meet the requirements of table I.

TABLE I. Type I and type II heat volume requirements.

Type	Size	Heat volume (seconds)
I	1	Less than 85
I	2	Less than 65
I	3	Less than 45
II	N/A	Less than 60

3.3.9 Heating time. The soldering guns shall reach a temperature sufficient to melt 1/16 inch diameter, 60/40 tin-lead solder within 15 seconds.

3.3.10 Efficiency. Input-output efficiency of the soldering guns shall be not less than 25 percent.

3.3.11 Reliability. The type I soldering guns shall be capable of operating under continuous cycles of 1 minute on and 4 minutes off for a period of 8 hours without failure. The type II soldering guns shall be capable of operating under continuous cycles of 10 minutes on and 10 minutes off for a period of 8 hours without failure. The type IV soldering guns shall operate continuously for a period of 8 hours without failure.

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3.3.12 Electrical requirements. Unless otherwise specified (see 6.2.1), the soldering guns shall be designed to operate on 120 volts ± 10 percent, AC, 60 Hz, single phase input power.

3.3.13 Power cord. The soldering guns shall be furnished with a power cord of not less than 4-1/2 feet in length. Unless otherwise specified (see 6.2.1), the type I and type II soldering guns shall be furnished with a three conductor power cord. The type IV soldering guns shall be furnished with a three conductor power cord only.

3.3.14 Tip temperature. The tip temperature shall not exceed 1300 degrees Fahrenheit (F) for the type I and type II soldering guns

3.3.15 Housing (case). The soldering gun housing shall be manufactured from thermosetting plastic or high temperature thermoplastic. The housing shall consist essentially of a pistol grip type handle and transformer housing or housing and mounting for the heating element as applicable.

3.3.16 Temperature rise of housing (case). The maximum temperature rise of any part of the housing with which the hand normally comes in contact, shall not exceed 50 degrees F above ambient temperature when the type I soldering guns are continuously cycled for 1 minute on and 4 minutes off and the type II and type IV are cycled for 30 seconds on and 30 seconds off for a 20 minute period. The front surface of the housing which surrounds the soldering tip shall not be construed as a part of the housing with which the hand normally comes in contact.

3.3.17 Handle. The handle shall be of a pistol-grip design providing a comfortable grip to the hand. The external surface of the handle shall be smooth or textured, without sharp edges or projections and shall be free of screw heads and other metallic projections. Heads of screws and other metallic projections shall be of a sufficient depth below the external surface to prevent the transmittal of heat to the operator's hand. The cord end of the handle shall be suitably flared and smoothly finished, or provided with other suitable means to prevent chafing of the power cord at its entrance to the handle. The handle shall contain a trigger switch positioned so that energy input to the soldering guns is controlled by the forefinger.

3.3.18 Trigger switch The soldering guns shall be furnished with a trigger switch. When a two position switch is furnished, the soldering guns shall be rated and tested at the highest heat volume.

3.3.19 Illumination. The type I, size 2 and 3, and the type II soldering guns shall be furnished with a lamp or lamps to satisfactorily illuminate the tip working area. The illuminating lamp or lamps shall be enclosed by the housing in such a manner that lamp breakage will not create an electrical hazard. The lamp shall be replaceable without requiring the use of special tools.

3.3.20 Replacement tips and heating units. When specified (see 6.2.1), replacement tips, heating elements, or combination tip and heating elements shall be the appropriate replacement part(s) for the soldering gun brand name and model specified by the procuring activity.

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3.4 Detailed requirements.

3.4.1 Type I soldering guns. The type I soldering guns shall conform to tables I and II.

TABLE II. Dimensions of type I soldering guns.

Size	Weight maximum 1/	Overall length maximum	Height maximum	Width maximum
1	30 oz	10 in	7 in	2-1/4 in
2	40 oz	11 in	7 in	2-1/4 in
3	50 oz	11 in	7 in	2-1/4 in

1/ Weight is without power cord.

3.4.1.1 Transformer. The transformer for the type I soldering guns shall be of the step-down voltage type securely enclosed in the housing portion of the plastic housing. The secondary of the transformer shall be connected to the soldering tip holder.

3.4.1.2 Tips. Tips for type I soldering guns shall be of the wire-type design so that the tip is part of the secondary circuit of the transformer and is the primary source of heat. Tips shall be made of commercially pure copper, tellurium copper alloy, leaded copper, or any other equally suitable material or combination of materials which may be readily tinned. The tip shall be secured to the secondary circuit with a suitable mechanical device so that the union of the tip and secondary windings shall conduct the secondary current effectively from the secondary winding to the higher resistance tip.

3.4.2 Type II soldering guns. The type II soldering guns shall conform to table I. The soldering guns shall be designed in such a manner that the tip and heater assembly may be replaced without the use of special tools.

3.4.2.1 Heater assembly . The heater assembly shall be manufactured from and insulated with materials which meet the requirements of the UL 499.

3.4.2.2 Soldering tips. The tips for the type II soldering guns shall be made of commercially pure copper, tellurium copper, lead copper, or steel. Unless otherwise specified (see 6.2.1), a pyramid shape tip shall be furnished with the soldering gun. When a different shaped tip is required, the shape shall be as specified.

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3.4.3 Type IV Soldering guns.

3.4.3.1 Type IV, Class 1. The type IV, class 1 soldering guns shall control temperature so the average tip temperature is within ± 25 degrees F of the preselected idle temperature and the tip temperature stability is ± 10 degrees Fahrenheit. The maximum leakage voltage when measured from the soldering gun tip to ground shall not exceed 0.002 volt RMS over a frequency range of 20 Hz to 20 kilo Hz, at operating temperature. Switching transients at the tip shall not exceed a value of 0.006 volt peak for more than 1 microsecond (see figures 1 and 2).

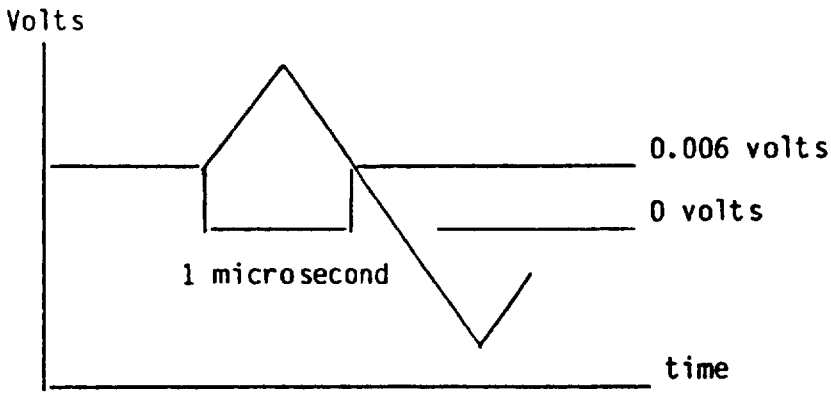


FIGURE 1. Acceptable switching transient waveform.

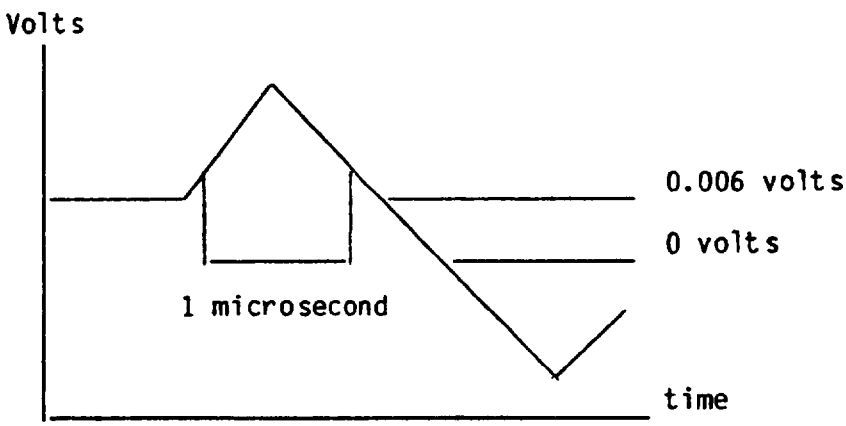


FIGURE 2. Unacceptable switching transient waveform.

3.4.3.2 Type IV, Class 2. The type IV, class 2 soldering guns shall control temperature so the average tip temperature is within ± 25 degrees F of the selected idle temperature and tip temperature stability is ± 10 degrees Fahrenheit. The gun shall have a controlled temperature, which is adjustable, from less than 500 degrees to greater than 800 degrees F. The temperature shall be adjustable with a calibrated dial. The maximum tip leakage voltage and switching transient voltage levels shall be in accordance with 3.4.3.1.

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3.4.4.1 Heater assembly. The type IV soldering guns heater assembly shall conform to 3.4.2.1.

3.4.4.2 Soldering tips. The tips for the type IV soldering guns shall be made of commercially pure copper, tellurium copper, or lead copper and then coated with iron. The iron coating shall be not less than 0.005 inch thick in the working area. The tips shall also be coated with 60/40 tin lead solder in the working area. Unless otherwise specified, a screwdriver shaped tip shall be furnished with the soldering gun. When a different shaped tip is required, the shape shall be as specified (see 6.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspections set forth in this inspection shall become a part of the contractor's overall inspection system or quality control program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.2. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6, and the inspection in 4.7. The soldering gun shall pass the examination, tests, and inspection to be accepted.

4.4 Quality conformance inspection. A quality conformance sampling inspection shall be applied to each item offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.3, and the

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inspection in 4.7. Failure of an item to pass the examination, and the inspection shall be cause for rejection.

4.4.1 Quality conformance sample. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105. Sampling shall be at level S-1 with an AQL of 4.0.

4.5 Examination. The first article and each soldering gun in the sample selected for quality conformance inspection shall be visually examined to determine compliance with 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5, 3.3.6, 3.3.13, 3.3.14, 3.3.15, 3.3.17, 3.3.18, 3.3.19, 3.4.1.1, 3.4.1.2, 3.4.2, 3.4.2.1, 3.4.2.2, and 3.4.4.2.

4.6 Tests.

4.6.1 Test conditions. All tests shall be conducted in an ambient temperature within the range of 75 to 80 degrees F with the soldering guns operating at their designed voltage.

4.6.2 Impact test. The soldering guns shall be tested for conformance with 3.3.7. The soldering guns shall be dropped from a height of three feet onto a hard surface. After this test, the soldering gun shall be examined for damage that would cause a safety hazard to the user.

4.6.3 Heat volume. The heat volume of the soldering guns shall be tested for conformance with 3.3.8. The heat volume test shall be accomplished in the following manner. A copper bar 0.125 by 0.5 by 1.5 inches in size with a tolerance of plus or minus 0.01 inches in all dimensions and weighing 12.9 grams plus or minus 0.9 grams shall be tinned on both ends. A 0.25 inch length of 0.0625 inch diameter 60/40 tin-lead solder shall be placed 0.25 inch from each end of the copper bar. The soldering gun tip's working surface which has been tinned shall be placed on top of one length of solder. Simultaneously the gun shall be energized and a timer started. The time required to melt the solder on the opposite end of the bar shall be in accordance with the heat volume requirements of table I for the applicable type and size soldering gun.

4.6.4 Heating time. The soldering guns shall be tested for conformance with the heating time requirements specified in 3.3.9. A one to two inch length of 0.0625 inch diameter 60/40 tin-lead solder shall be placed at a 90 degree angle and in contact with the soldering tip. The soldering gun shall then be energized. The time required for the first drop of solder to form shall not exceed 15 seconds.

4.6.5 Efficiency. The soldering guns shall be tested to determine compliance with the requirements specified in 3.3.10.

4.6.5.1 Efficiency test equipment. The following equipment is required to perform the soldering gun efficiency test. A 80 milliliter (mL) glass beaker, a low mass temperature probe (range 20° Centigrade (C) to 100° C $+1^{o}$ C), or equal temperature measuring device, distilled water, a wattmeter (0 to 300 watt $\pm$ 1%), and a stopwatch or timer capable of measuring to one second for a two minute period. A low range watthour meter may be used in place of the wattmeter and stopwatch.

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4.6.5.2 Efficiency test procedure. The 80 mL beaker shall be filled with 60 mL of distilled water. The temperature probe shall be inserted into the beaker of water and allowed to stabilize at ambient temperature. The soldering gun shall be plugged into the wattmeter and the applied voltage adjusted to the nominal line voltage as listed on the soldering gun. The ambient water temperature shall be recorded (T1). The soldering gun shall be energized and after 15 seconds, the tip shall be submerged into the beaker of water. Simultaneously the timing device shall be started. The barrel portion of the type II and type IV guns shall not be submerged. When the wattage reading stabilizes, it shall be recorded (W). When 120 seconds have elapsed on the timing device, the gun shall be turned off. The tip shall remain immersed until the water temperature rises to a maximum. The maximum temperature shall be recorded (T2). The efficiency shall be calculated from the following formula. The efficiency shall be 25 percent or greater.

$$\frac{(T2 - T1) \times 60}{(W \times 120)^{*} \times (239 \times 10\text{-}3)} = \frac{\text{Calories (Output)}}{\text{Calories (Input)}} = \text{Efficiency}$$

T1 - Start temperature of water
T2 - Maximum temperature of water
W - Wattage
239 x 10-3 = Calories/watt second.

*Watthour meter reading in watt seconds may be substituted.

4.6.5.2.1 Efficiency error. If this test does not load controlled type soldering guns below their controlled temperature and result in full on power conditions, an efficiency error will result and a fast response low range watthour meter must be used for a correct efficiency test.

4.6.6 Reliability test. The soldering guns shall be tested for conformance with 3.3.11. All tip-to-tip holder connections shall be kept tight to eliminate an increase in contact resistance. Wattage reading shall be taken every two hours. Tips and tip assemblies shall be considered to have failed when they become open or the input wattage changes in excess of 15 percent of the initial value.

4.6.7 Tip temperature. The type I and type II soldering gun tip temperature shall be measured for conformance with 3.3.14. The temperature shall be measured by attaching a thermocouple to the center of the wetted surface of the tip and measuring the voltage generated by the thermocouple or by the use of other suitable means. The temperature shall be measured after the soldering gun has been energized for one minute.

4.6.8 Temperature rise of housing (case). The temperature rise of the soldering gun housing (case) shall be measured for conformance with 3.3.16. The temperature rise shall be measured by any suitable means which is accurate to $\pm$ 3 degrees Fahrenheit.

4.6.9 Temperature control test. The temperature control of the type IV, class 1 soldering guns shall be tested for conformance with 3.4.3.1 and the type IV, class 2 soldering guns shall be tested for conformance with 3.4.3.2. The soldering guns may be tested in accordance with paragraph 4.6.9.1 or paragraph 4.6.9.2.

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4.6.9.1 Temperature control test procedure, wedged thermocouple method.

4.6.9.1.1 Equipment required for temperature control test. The following equipment is required to perform the temperature control test using an embeded thermocouple. A thermocouple and a device capable of measuring the thermocouple voltage with a measurement accuracy of +_ 0.5 degree Fahrenheit.

4.6.9.1.2 Temperature control test procedure. The temperature test of the soldering guns under test shall be accomplished as follows:

a. Drill a hole approximately 0.052 inch in diameter and 1/8 inch deep in the tip of the soldering gun under test. Ensure no molten solder enters the hole.

b. Insert the twisted thermocouple so the last twist is exactly at the edge of the hole. Wedge a small piece of copper into the hole to hold the thermocouple in tight contact with the tip. Trim off any excess from the wedge.

c. Attach the thermocouple wires to the voltage measuring device and energize the gun under test. The type IV, class 1, shall be tested at rated temperature and the type IV, class 2, shall be tested at 500 and 800 degrees Fahrenheit. Allow the gun to stabilize at idle temperature. Take a series of at least 5 temperature readings at intervals of one minute apart. Calculate the average temperature.

d. The average temperature shall meet the requirements of 3.4.3.1 for the type IV, class 1 and 3.4.3.2 for the type IV, class 2 soldering guns.

4.6.9.2 Temperature control test procedure, welded thermocouple method.

4.6.9.2.1 Equipment required for test. The following equipment is required to perform the temperature control test using a welded thermocouple. A device capable of measuring the thermocouple voltage with measurement accuracy of + 0.5 degree F, fume hood, safety goggles, plastic beaker, miniature welder, and air supply.

4.6.9.2.2 Materals required for test. The following materials are required to perform the temperature control test. Concentrated flouroboric acid, hydrogen peroxide (3% solution), 30 gage iron-constantan thermocouple wire with glass insulation, running water, and 20 to 30 gage bare wire.

4.6.9.2.3 Temperature control test procedure. The temperature test of the soldering guns under test shall be accomplished as follows:

a. Mix concentrated flouroboric acid and hydrogen peroxide 1:1 by volume in plastic beaker.

b. Cut thermocouple wire to desired length and remove 1/2 inch of insulation from each end. Clean both ends of thermocouple wire with concentrated flouroboric acid and hydrogen peroxide mixture. Rinse each end of thermocouple wire in running water. Cross two wires on one end of thermocouple and weld together forming a welded joint.

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c. File one section in working area of the soldering gun tip to expose iron plating. Clean working area of tip with concentrated flouroboric acid and hydrogen peroxide mixture and rinse with running water. Blow tip dry with air. Place tip in gun and secure in place.

d. Position welded junction of thermocouple junction in middle of cleaned working surface on tip and weld in place. Temperature measurement is to be made in the working area of tip, if working area is not obvious, weld thermocouple to cleaned area 0.06 inch ±0.01 inch from end of tip.

e. Attach the thermocouple wires to the voltage measuring device and energize the gun under test. The type IV, class 1, shall be tested at rated temperature and the type IV, class 2, shall be tested at 500 and 800 degrees Fahrenheit. Allow the gun to stabilize at idle temperature. Take a series of at least 5 temperature readings at intervals of one minute apart. Calculate the average temperature.

f. The average temperature shall meet the requirements of 3.4.3.1 for the type IV, class 1 and 3.4.3.2 for the type IV, class 2 soldering guns.

4.6.10 Leakage voltage test. The leakage voltage requirements of the type IV, class 1 soldering guns shall be tested for conformance with 3.4.3.1 and the type IV, class 2 soldering shall be tested for conformance with 3.4.3.2.

4.6.10.1 Equipment required for leakage voltage test. The following equipment is required to perform the leakage voltage test. An AC voltmeter capable of indicating 0.002 volt over the frequency range specified in 3.4.3 to an accuracy of ± 3 percent with an input impedance of 10 mega ohms or greater, miscellaneous shielded cables and clips, and a test plate manufactured from a piece of brass or copper shim stock approximately 1.5 inches long by 0.75 inch wide and from 0.008 to 0.020 inch thick.

4.6.10.2 Leakage test procedure. The test for voltage leakage shall be accomplished as follows:

a. Remove any oxidation from the shim stock.

b. Bond a small pool of solder to the shim stock using sufficient heat to insure a well wetted bond between the shim stock and solder.

c. If the voltmeter is not battery operated, it must be plugged into the same duplex wall receptacle as the unit under test.

d. Attach a cable between the test plate and ground pin of the soldering gun under test.

e. Attach the positive and negative cable of the voltmeter to the test plate (see figure 9). Ensure that a good electrical connection is made.

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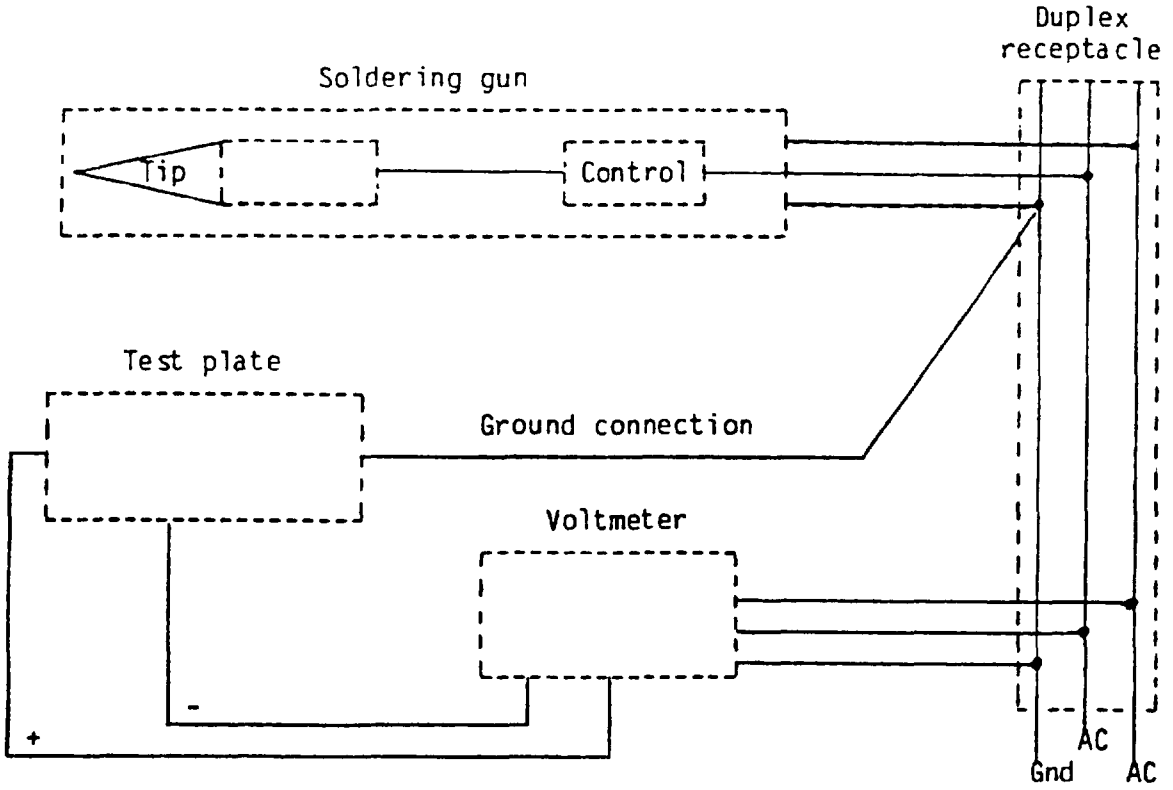


FIGURE 9. Measurement of test system voltage level.

f. Apply power to the voltmeter and soldering gun under test. Adjust type IV, class 2 soldering guns to 800 degrees Fahrenheit. Allow the soldering gun to reach idle temperature. Select the proper range on the voltmeter and measure the measurement system voltage. Record as V1.

g. Remove the cable connected between the test plate and the ground of the soldering gun under test. Remove the voltmeter negative cable from the test plate and attach to the soldering gun under test ground (see fig 10).

h. Place the tip of the soldering gun in the solder to melt. Allow temperature to stabilize and then measure voltage. Record as V2.

i. The leakage voltage of the soldering gun under test is the difference of V2 and V1. This voltage shall be 0.002 volt or less.

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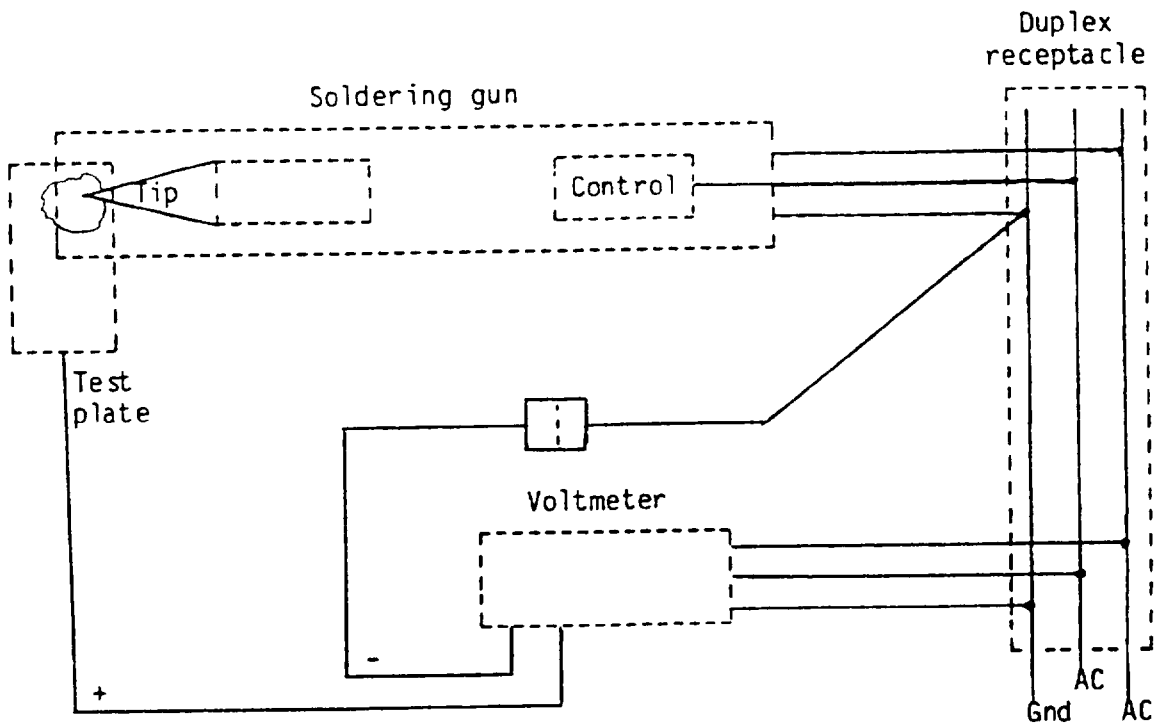


FIGURE 10. Measurement of test system voltage level.

4.6.10.3 Transient voltage test. The transient voltage requirements of the type IV, class 1 and 2 soldering guns shall be tested for conformance with 3.4.3. The test shall be accomplished with the class 1 guns operating at rated temperature and the class 2 guns operating at 800 degrees F and with the ground connected.

4.7 Packaging inspection. Packaging of the soldering gun shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The soldering guns covered by this specification are intended to be used for intermittent soldering where rapid availability for use is required without the continuous consumption of electric current.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, size or temperature, quantity of soldering guns required (see 1.2).

c. First article, if required (see 3.2 and 4.3).

d. Electrical input requirements, if different (see 3.3.12).

e. Type I and II soldering guns power cord, if different (see 3.3.13).

f. Brand name and model number of soldering gun which replacement tip and/or heating unit will be used on (see 3.3.20).

g. Shape of tip for type II guns, if different (see 3.4.2.2).

h. Shape of tip for type IV guns, if different (see 3.4.4.2).

i. First article inspection, if different (see 4.3).

j. Quality conformance inspection, if different (see 4.4).

k. If requirement for packing, packaging,and marking in accordance with MIL-W-45562 is required, specify level (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Measurement system. In this specification all dimensions are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

6.5 Part or identifying number (PIN). The following part or identifying numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

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-- Temperature:	0 - Not applicable	
	1 - 600 degrees	
	2 - 700 degrees	
---- Size:	0 - Not applicable	
	1 - Less than 85 seconds	
	2 - Less than 65 seconds	
	3 - Less than 45 seconds	
------ Class:	A - Not applicable	
	B - Class 1 - Fixed temperature	
	C - Class 2 - Variable temperature	
-------- Type:	1 - Type I	- Soldering gun, wire tip, with transformer.
	2 - Type II	- Soldering gun, solid tip
	3 - Type IV	- Soldering gun, solid tip, temperature controlled
---------- Federal specification designator		

6.6 Cross reference. Cross reference of classification changes between this specification and the preceding issue are as follows:

W-S-564A	W-S-564B	W-S-564C
Type I	Type I	Type I
	Size 1	Size 1
	Size 2	Size 2
	Size 3	Size 3
Type II	Type II	TypeII
	Size 1	Deleted
	Size 2	Deleted
	Size 3	Deleted
	Type III (New type)	(Deleted)
		Type IV (New type)
		Class 1 (New class)
		Temperature 600
		Temperature 700
		Class 2 (New class)
Cord Assembly A	Deleted	
Cord Assembly B	Deleted	
Class 1	Deleted	
Class 2	Deleted	
Class 3	Deleted	

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MILITARY INTEREST

Custodians:

Army - AL
Air Force - 99

Review activities:

Air Force - 84
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

DLA-IP

DOD project 3439-0753

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